Chapter 8. Milnor's seventeen problems, with solutions

Milnor, "Topology from the Differentiable Viewpoint", §8, consists of seventeen problems "for the reader". Here they are, each with a complete solution in the style of the preceding chapters. A reader who has followed the book should try each one before reading on; the solutions are arranged so that the statement comes first and the argument after a rule.

Problem 1. The degree of a composition

Show that $\deg(g \circ f) = \deg g \cdot \deg f$.

Let $f : M \to N$, $g : N \to P$ be smooth maps between compact connected oriented manifolds of the same dimension. The regular values of g form an open dense subset of P (Chapter 5, proof of Theorem B), and so do those of $g \circ f$; pick z in both. For $y \in g^{-1}(z)$ and $x \in f^{-1}(y)$, $d(g \circ f)_x = dg_y \circ df_x$ is invertible and dg_y is invertible, so df_x is invertible: every $y \in g^{-1}(z)$ is a regular value of f. Since $(g \circ f)^{-1}(z)$ is the disjoint union of the $f^{-1}(y)$ over $y \in g^{-1}(z)$, and signs multiply,

$$\deg(g\circ f) = \sum_{y\in g^{-1}(z)} \sum_{x\in f^{-1}(y)} \operatorname{sign} dg_y \operatorname{sign} df_x = \sum_{y\in g^{-1}(z)} \operatorname{sign} dg_y \cdot \deg(f; y) = \deg f \sum_{y\in g^{-1}(z)} \operatorname{sign} dg_y = \deg f$$

Problem 2. Polynomials have degree n

Show that a complex polynomial of degree n gives a smooth map of the Gauss sphere S^2 to itself of degree n.

Chapter 1, §8, made P into a smooth map f of the Riemann sphere $\widehat{\mathbb{C}}$. Orient $\widehat{\mathbb{C}}$ by the chart χ_0 (the standard orientation of $\mathbb{C} = \mathbb{R}^2$); the change of coordinates to χ_∞ is $z \mapsto 1/z$, holomorphic with nonzero derivative, and multiplication by a nonzero complex number $w = a + bi$ is the real matrix $\begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ with determinant $|w|^2 > 0$, so holomorphic maps preserve orientation and the two charts orient $\widehat{\mathbb{C}}$ consistently. For a regular value $y \notin B$ (Chapter 1, Claim 8.5), the equation $P(z) = y$ has n roots counted with multiplicity, all simple because $P'(z) \neq 0$ at each root (y is not a critical value), so n distinct preimages, each with \$df_z = \$ multiplication by $P'(z) \neq 0$, of sign $+1$. Hence $\deg f = n$. $\square$

Problem 3. Maps that are never antipodal are homotopic

If $f, g : X \to S^p$ satisfy $\|f(x) - g(x)\| < 2$ for all x, then $f \simeq g$, smoothly if f, g are smooth.

The hypothesis says $g(x) \neq -f(x)$. Then $(1-t)f(x)+t\,g(x) \neq 0$ for $t \in [0,1]$: if it vanished, taking norms would give $(1-t) = t$, so $t = \frac{1}{2}$ and $f(x) = -g(x)$. Hence

$$F(x,t) := \frac{(1-t)f(x) + t\,g(x)}{\|(1-t)f(x) + t\,g(x)\|}$$

is a well-defined homotopy $X \times [0,1] \to S^p$ from f to g, smooth when f, g are. □

Problem 4. Smoothing maps and homotopies

If X is compact, every continuous map $X \to S^p$ is uniformly approximable by smooth maps, and continuously homotopic smooth maps are smoothly homotopic.

Let $X \subseteq \mathbb{R}^k$ be compact and $G : X \to S^p \subseteq \mathbb{R}^{p+1}$ continuous. By the Weierstrass approximation theorem applied to each component, there is a polynomial map $Q : \mathbb{R}^k \to \mathbb{R}^{p+1}$ with $\|Q(x) - G(x)\| < \varepsilon$ on X, for any given $\varepsilon < 1$. Then $\|Q(x)\| > 1-\varepsilon > 0$, so $\tilde{G} := Q/\|Q\|$ is smooth on a neighbourhood of X, maps into S^p, and $\|\tilde{G}(x) - G(x)\| \leq \|\tilde{G}(x) - Q(x)\| + \|Q(x) - G(x)\| < \varepsilon + \varepsilon$. For the second statement, let $H : X \times [0,1] \to S^p$ be a continuous homotopy between smooth f and g. Apply the first part to H on the compact set $X \times [0,1] \subseteq \mathbb{R}^{k+1}$ with $\varepsilon = \frac{1}{4}$: a smooth $\tilde{H}$ with $\|\tilde{H} - H\| < \frac{1}{2}$. Then $\tilde{H}(\cdot, 0)$ is smooth and within $\frac{1}{2} < 2$ of f, so smoothly homotopic to f by Problem 3; likewise $\tilde{H}(\cdot, 1) \simeq g$; and $\tilde{H}$ is a smooth homotopy between them. Concatenate (Chapter 4, Claim 1.1). □

Problem 5. Maps into a higher-dimensional sphere are null-homotopic

If $m < p$, every map $M^m \to S^p$ is homotopic to a constant.

By Problem 4 we may assume f smooth. Since $m < p$, every point of M is a critical point, so $f(M)$ is the set of critical values, which has measure zero by Sard's theorem (via charts, as in Brown's theorem); in particular f is not onto. Choose $y \notin f(M)$ and let $h : S^p \setminus \{y\} \to \mathbb{R}^p$ be stereographic projection. Then $F(x,t) := h^{-1}\big((1-t)\,h(f(x))\big)$ is a smooth homotopy from f to the constant map $h^{-1}(0)$. □

Problem 6. Brouwer's fixed point theorem for spheres

A map $f : S^n \to S^n$ with $\deg f \neq (-1)^{n+1}$ has a fixed point.

If $f(x) \neq x$ for all x, then $f(x) \neq -\alpha(x)$ where $\alpha(x) = -x$, so $\|f(x) - \alpha(x)\| < 2$ and $f \simeq \alpha$ by Problem 3. Hence $\deg f = \deg \alpha = (-1)^{n+1}$ (Chapter 5, §5), contrary to hypothesis. □

Problem 7. Maps of odd degree carry some antipodal pair to an antipodal pair

A map $f : S^n \to S^n$ of odd degree carries some pair $\{x, -x\}$ to a pair $\{y, -y\}$.

Suppose $f(-x) \neq -f(x)$ for all x. Then $f(x) + f(-x) \neq 0$, and

$$h(x) := \frac{f(x) + f(-x)}{\|f(x) + f(-x)\|}$$

is a smooth map $S^n \to S^n$ with $h(-x) = h(x)$. It is homotopic to f by Problem 3: $h(x) = -f(x)$ would mean $f(x) + f(-x) = -\lambda f(x)$ with $\lambda > 0$, so $f(-x) = -(1+\lambda)f(x)$, impossible for unit vectors. So $\deg h = \deg f$. But an even map has even degree: let y be a regular value of h; the preimage $h^{-1}(y)$ is invariant under α and contains no fixed point of α, so it splits into pairs $\{x, -x\}$. Differentiating $h = h \circ \alpha$ gives $dh_x = dh_{-x} \circ d\alpha_x$, so $\operatorname{sign} dh_x = \operatorname{sign} dh_{-x} \cdot (-1)^{n+1}$: each pair contributes ± 2 or 0 to the degree. Hence $\deg h$ is even, and so is $\deg f$. □

Problem 8. Tangent spaces of products

For $M \subseteq \mathbb{R}^k$, $N \subseteq \mathbb{R}^l$, show $T(M \times N)_{(x,y)} = TM_x \times TN_y$.

If $g : U \to M$ and $h : V \to N$ are parametrizations around x and y, then $g \times h : U \times V \to M \times N$ is a parametrization around (x, y) (smooth, injective, with inverse (g^{-1}, h^{-1}) locally extended), so $M \times N$ is a manifold of dimension $m+n$, and $d(g \times h)_{(u,v)} = dg_u \times dh_v$ has image $\operatorname{im} dg_u \times \operatorname{im} dh_v = TM_x \times TN_y$. □

Problem 9. The graph of a smooth map

The graph $\Gamma = \{(x, f(x))\}$ of $f : M \to N$ is a smooth manifold with $T\Gamma_{(x,y)} =$ the graph of df_x.

The map $\gamma(x) := (x, f(x))$ is smooth $M \to M \times N$, injective, with inverse the projection $\Gamma \to M$, which is the restriction of a linear map and hence smooth. So γ is a diffeomorphism $M \to \Gamma$, and composing it with parametrizations of M gives parametrizations of Γ: Γ is an m-manifold. Its tangent space is the image of $d\gamma_x = (\mathrm{id}, df_x)$, namely $\{(v, df_x v) : v \in TM_x\}$, the graph of df_x inside $TM_x \times TN_y$. □

Problem 10. The tangent bundle

$TM = \{(x, v) \in M \times \mathbb{R}^k : v \in TM_x\}$ is a smooth manifold, and $f \mapsto df$ is functorial.

For a parametrization $g : U \to M$, define $\tilde{g} : U \times \mathbb{R}^m \to TM$, $\tilde{g}(u, w) := (g(u), dg_u(w))$. It is smooth, injective (g is, and dg_u is injective by Chapter

1, Claim 3.2), and its image is $\{(x,v) : x \in g(U)\}$, open in TM. Its inverse is smooth: $x \mapsto g^{-1}(x)$ is smooth (locally the restriction of a smooth F with $F \circ g = \mathrm{id}$), and $w = dF_x(v)$ because $dF_x \circ dg_u = \mathrm{id}$. So the $\tilde{g}$ are parametrizations and TM is a $2m$-manifold. For smooth $f : M \to N$, the map $df : TM \to TN$, $(x,v) \mapsto (f(x), df_x v)$, reads in these parametrizations as $(u,w) \mapsto (\varphi(u), d\varphi_u w)$ with $\varphi = h^{-1} \circ f \circ g$, which is smooth. The identities $d(\mathrm{id}) = \mathrm{id}$ and $d(g \circ f) = dg \circ df$ are the chain rule, pointwise. □

Problem 11. The normal bundle and the ε-neighbourhood

$E = \{(x,v) : v \perp TM_x\}$ is a smooth manifold; for compact M, $(x,v) \mapsto x + v$ maps the ε-neighbourhood of $M \times 0$ in E diffeomorphically onto N_ε.

This is Chapter 6, Proposition 4.1, where it was proved in full for use in the Poincaré–Hopf theorem. □

Problem 12. The closest-point retraction, and smoothing maps into a manifold

$r(x+v) = x$ is the closest point of M; using r, prove Problem 4 with S^p replaced by a compact manifold M.

The first statement is Chapter 6, Proposition 4.1 (i). For the second: let X be compact and $G : X \to M \subseteq \mathbb{R}^k$ continuous. Choose ε as in Proposition 4.1 and a polynomial Q with $\|Q - G\| < \varepsilon$ on X (Weierstrass). Then $Q(X) \subseteq N_\varepsilon$, and $\tilde{G} := r \circ Q$ is smooth with values in M and $\|\tilde{G}(x) - G(x)\| \le \|r(Q(x)) - Q(x)\| + \|Q(x) - G(x)\| < 2\varepsilon$, because $r(Q(x))$ is the closest point of M to $Q(x)$ and $G(x) \in M$ is within ε. For homotopies: approximate a continuous homotopy H by a smooth $\tilde{H} = r \circ Q$ within ε; for smooth f within ε of $\tilde{H}(\cdot, 0)$, the segment $(1-t)f(x) + t\tilde{H}(x,0)$ stays within ε of $f(x) \in M$, so r of it is a smooth homotopy in M. Concatenate. □

Problem 13. Linking numbers

For disjoint compact oriented boundaryless $M^m, N^n \subseteq \mathbb{R}^{k+1}$ with $m + n = k$, $\lambda(x,y) = (x-y)/\|x-y\|$ has degree $l(M,N)$. Prove $l(N,M) = (-1)^{(m+1)(n+1)} l(M,N)$; prove $l(M,N) = 0$ if M bounds an oriented X disjoint from N; define linking in S^{m+n+1}.

Symmetry. Let $\sigma : N \times M \to M \times N$ swap the factors and α the antipodal map of S^k. Then $\lambda_{N,M} = \alpha \circ \lambda_{M,N} \circ \sigma$. With product orientations, σ has degree $(-1)^{mn}$ (moving n vectors past m vectors in an ordered basis is mn transpositions), and $\deg \alpha = (-1)^{k+1} = (-1)^{m+n+1}$ (Chapter 5). By Problem 1, $l(N,M) = (-1)^{mn+m+n+1} l(M,N) = (-1)^{(m+1)(n+1)} l(M,N)$.

Bounding. If $M = \partial X$ with X compact oriented and $X \cap N = \emptyset$, then λ extends to $X \times N \to S^k$ by the same formula, and $\partial(X \times N) = M \times N$ with the product orientation (the outward normal lies in the X factor, which comes first). By Chapter 5, Lemma 3.1, $\deg \lambda = 0$.

In the sphere. For disjoint $M, N \subseteq S^{m+n+1}$, the union $M \cup N$ has measure zero, so there is a point q outside it; stereographic projection h_q from q carries M, N to disjoint submanifolds of $\mathbb{R}^{m+n+1}$, and one defines $l(M,N) := l(h_q(M), h_q(N))$. It does not depend on q: two admissible points q, q' can be joined by a path q_t in the complement of $M \cup N$ (the complement of a compact set of dimension $\leq \max(m,n) \leq m+n-1$, codimension at least 2, is connected: a path between q and q' can be perturbed off it by Sard's theorem applied to the map (parameter, perturbation) $\mapsto$ sphere), and then $(x,y,t) \mapsto \lambda(h_{q_t}(x), h_{q_t}(y))$ is a homotopy, so the degree is constant by Chapter 5, Theorem B. $\square$

Problem 14. The Hopf invariant

For $f : S^{2p-1} \to S^p$ and regular values $y \neq z$, the manifolds $f^{-1}(y)$, $f^{-1}(z)$ are oriented as in Chapter 5 and $l(f^{-1}(y), f^{-1}(z))$ is defined. Prove: (a) it is locally constant in y; (b) if $\|f-g\| < \|y-z\|$ pointwise and y, z are regular for g too, then $l(f^{-1}(y), f^{-1}(z)) = l(g^{-1}(y), f^{-1}(z)) = l(g^{-1}(y), g^{-1}(z))$; (c) it depends only on the homotopy class of f, not on y, z. This integer $H(f)$ is the Hopf invariant.

Orientation. $f^{-1}(y)$ has dimension $2p-1-p = p-1$, and $(p-1)+(p-1)+1 = 2p-1$, so linking numbers in S^{2p-1} make sense. Orient $f^{-1}(y)$ so that a basis of its tangent space followed by the framing f^*v is a positive basis of $T(S^{2p-1})$; this is the orientation Chapter 5, Lemma 3.1, used for the arcs of $F^{-1}(y)$.

(a) Let y' be close to y, with the short arc from y to y' consisting of regular values and avoiding z. Chapter 7, Lemma 2.2, gives rotations r_t and the manifold $X = F^{-1}(y') \subseteq S^{2p-1} \times [0,1]$, $F(x,t) = r_t f(x)$, whose slice at height t is $N_t := f^{-1}(r_t^{-1} y')$, with $N_0 = f^{-1}(y')$ and $N_1 = f^{-1}(y)$; the framing F^*v orients X compatibly with the orientations of N_0 and N_1 (up to the boundary sign of Chapter 5, which is the same at both ends of the comparison below). Every N_t is disjoint from $f^{-1}(z)$ because $r_t^{-1} y' \neq z$. Choose $q \in S^{2p-1}$ outside the measure-zero set $\bigcup_t N_t \cup f^{-1}(z)$ and project stereographically. The map $\Lambda : X \times f^{-1}(z) \to S^{2p-2}$, $((x,t), x') \mapsto \lambda(h_q x, h_q x')$, is smooth and defined everywhere, and $\partial(X \times f^{-1}(z)) = (N_1 - N_0) \times f^{-1}(z)$ as oriented manifolds. By Chapter 5, Lemma 3.1, $\deg \Lambda|_\partial = 0$, i.e. $l(f^{-1}(y), f^{-1}(z)) = l(f^{-1}(y'), f^{-1}(z))$.

(b) Measure distances along the sphere by angle; the hypothesis $\|f-g\| < \|y-z\|$ becomes $d(f(x), g(x)) \leq d(y,z) - \delta$ for some $\delta > 0$ (compactness). Let $F_t(x)$ be the point at fraction t along the short great-circle arc from $f(x)$ to $g(x)$; F is a smooth homotopy with $d(F_t(x), f(x)) \leq d(y,z) - \delta$ (reparametrize to be constant near $t = 0, 1$). Choose a regular value y' of F with $d(y', y) < \delta/2$;

then y' is regular for f and g (Chapter 7, Lemma 2.3), and $X := F^{-1}(y')$ is disjoint from $f^{-1}(z) \times [0,1]$: if $f(x) = z$ then $d(F_t(x), z) \leq d(y,z) - \delta < d(y', z)$, so $F_t(x) \neq y'$. Now the argument of (a), with this X, gives $l(f^{-1}(y'), f^{-1}(z)) = l(g^{-1}(y'), f^{-1}(z))$; combined with (a) for f and for g (moving y' back to y), $l(f^{-1}(y), f^{-1}(z)) = l(g^{-1}(y), f^{-1}(z))$. The second equality is the same argument with the roles of the two manifolds exchanged (Problem 13's symmetry shows nothing is lost by exchanging them).

(c) Independence of y, z. For a rotation r of S^p close to the identity, $\|r \circ f - f\| < \|y - z\|$, so by (b), $l(f^{-1}(y), f^{-1}(z)) = l((rf)^{-1}(y), (rf)^{-1}(z)) = l(f^{-1}(r^{-1}y), f^{-1}(r^{-1}z))$: the pair (y, z) can be moved by small rotations without changing the linking number, and by (a) y (and, symmetrically, z) can be moved alone within regular values. Since the regular values are open and dense and rotations act transitively on pairs at a given angle, any two admissible pairs are connected by such moves. *Homotopy invariance.* Given a smooth homotopy F_t from f to g, uniform continuity gives $0 = t_0 < \cdots < t_r = 1$ with $\|F_{t_i} - F_{t_{i+1}}\| < 1$; for each i choose common regular values y_i, z_i of F_{t_i} and $F_{t_{i+1}}$ with $\|y_i - z_i\| > 1$ (possible: regular values are open dense, and nearly antipodal pairs exist), and apply (b). □

Problem 15. Properties of the Hopf invariant, and the Hopf fibration

If p is odd, $H(f) = 0$. For $S^{2p-1} \xrightarrow{f} S^p \xrightarrow{g} S^p$, $H(g \circ f) = (\deg g)^2 H(f)$. The Hopf fibration $\pi(z_1, z_2) = h^{-1}(z_1/z_2)$ has $H(\pi) = 1$.

Odd p. By Problem 13 with $m = n = p - 1$, $l(f^{-1}(z), f^{-1}(y)) = (-1)^{p^2} l(f^{-1}(y), f^{-1}(z)) = -H(f)$ for p odd; by (c) the left side is also $H(f)$. So $H(f) = 0$.

Composition. Choose y, z regular for g and for $g \circ f$. Then $(g \circ f)^{-1}(y) = \bigsqcup_{y_i \in g^{-1}(y)} f^{-1}(y_i)$, each piece carrying its orientation multiplied by $\operatorname{sign} dg_{y_i}$ (the framing $(g \circ f)^* v = f^*(dg^{-1} v)$ is positive or negative at y_i accordingly), and similarly for z. The linking number is additive in each variable over disjoint unions and changes sign with orientation, so

$$H(g \circ f) = \sum_{i,j} \operatorname{sign} dg_{y_i} \operatorname{sign} dg_{z_j} \, l(f^{-1}(y_i), f^{-1}(z_j)) = \Big(\sum_i \operatorname{sign} dg_{y_i}\Big)\Big(\sum_j \operatorname{sign} dg_{z_j}\Big) H(f) = (\deg g)^2 H$$

using (c) for each pair (y_i, z_j).

The Hopf fibration. Write points of $S^3 \subseteq \mathbb{C}^2$ as (z_1, z_2) with $|z_1|^2 + |z_2|^2 = 1$, and $\pi(z_1, z_2) = h^{-1}(z_1/z_2) \in S^2$, with $\pi = h^{-1}(\infty)$ when $z_2 = 0$. The fibre over $h^{-1}(c)$ is the great circle $\{z_1 = c z_2\} \cap S^3$, and every value is regular (π is a submersion: in the chart $c = z_1/z_2$ its derivative has rank 2). Take $y = h^{-1}(1)$ and $z = h^{-1}(-1)$: the fibres are $C_+ = \{(w, w)/\sqrt{2} : |w| = 1\}$

and $C_- = \{(w,-w)/\sqrt{2}\}$. Project stereographically from $q = (0,i) \in S^3$, which lies on neither: $(x_1,x_2,x_3,x_4) \mapsto (x_1,x_2,x_3)/(1-x_4)$. With $w = e^{i\varphi}$,

$$C_+ \mapsto \frac{(\cos\varphi, \sin\varphi, \cos\varphi)}{\sqrt{2}-\sin\varphi}, \qquad C_- \mapsto \frac{(\cos\varphi, \sin\varphi, -\cos\varphi)}{\sqrt{2}+\sin\varphi},$$

circles in the planes $\{x_1 = x_3\}$ and $\{x_1 = -x_3\}$ respectively. Compute the degree of the linking map $\lambda(a,b) = (a-b)/\|a-b\|$ at the value $-e_2$: $a-b \parallel e_2$ forces a, b on the common line $x_1 = x_3 = 0$ of the two planes, where $\cos\varphi = 0$; C_+ meets it at $x_2 = \sqrt{2}+1$ and $x_2 = -(\sqrt{2}-1)$, C_- at $x_2 = \sqrt{2}-1$ and $x_2 = -(\sqrt{2}+1)$. Of the four pairs, exactly one has $a-b$ pointing in the direction $-e_2$: $a = (0,-(\sqrt{2}-1),0)$, $b = (0,\sqrt{2}-1,0)$. At this pair the tangent to C_+ is parallel to $(1,0,1)$ and the tangent to C_- to $(-1,0,1)$ (differentiate the parametrizations at $\varphi = -\pi/2$ and $\varphi = \pi/2$; the denominators have zero derivative there), so $d(a-b)$ maps onto the plane spanned by $(1,0,1)$ and $(-1,0,1)$, the tangent plane of S^2 at $-e_2$: the value $-e_2$ is regular with a single preimage, and $H(\pi) = \deg\lambda = \pm 1$. The sign needs care. With the book's own conventions (spheres oriented as boundaries of balls, the fibres oriented as preimages as in Chapter 5, $C_+ \times C_-$ oriented as a product, and λ as above) it is -1, not the $+1$ of the problem statement: the machine-checked edition carries out the determinant computation at the pair (a,b) and records this value (V8.15 there). The sign flips under any one of three equally natural changes of convention: $\lambda(a,b) = (b-a)/\|b-a\|$, the product taken in the order $C_- \times C_+$, or the target sphere of λ oriented the other way; the book's $+1$ corresponds to one of these. Reversing the orientation of S^3, or of the target S^2 of π, reverses both fibres at once and leaves $H(\pi)$ unchanged, and the order of the vectors in the preimage orientation (Milnor's ordering of the framing) makes no difference here. The important fact is $|H(\pi)| = 1$: the Hopf map is not null-homotopic, although $\pi_3(S^2)$ is invisible to the degree. □

Problem 16. Transversality

Submanifolds $N, N' \subseteq M$ intersect transversally if $TN_x + TN'_x = TM_x$ at every $x \in N \cap N'$. If N is framed, it can be deformed slightly to meet a given N' transversally, and the intersection is then a manifold.

Let $g : N \times \mathbb{R}^p \to V$ be a product neighbourhood (Chapter 7, Theorem 3.1) and $\pi : V \to \mathbb{R}^p$ the projection. For $u \in \mathbb{R}^p$ let $N_u := g(N \times \{u\}) = \pi^{-1}(u)$, a framed submanifold isotopic to $N = N_0$ through N_{su}, $s \in [0,1]$; for small u it is a small deformation of N. Consider the smooth map $\pi|_{N' \cap V} : N' \cap V \to \mathbb{R}^p$. **Claim:** N_u meets N' transversally iff u is a regular value of $\pi|_{N' \cap V}$. Indeed, at $x \in N_u \cap N'$, $d\pi_x : TM_x \to \mathbb{R}^p$ is onto with kernel $T(N_u)_x$, so $TN_u + TN'_x = TM_x$ iff $d\pi_x(TN'_x) = \mathbb{R}^p$, which is regularity of x for $\pi|_{N'}$. By Sard's theorem, regular values of $\pi|_{N' \cap V}$ are dense, so there are arbitrarily small u for which N_u is transverse to N', and then $N_u \cap N' = (\pi|_{N' \cap V})^{-1}(u)$ is a manifold

of dimension $\dim N' - p = n + n' - m$ by Chapter 2, Lemma 2.1 (empty if $n + n' < m$). $\square$

Problem 17. Products and the cohomotopy group

Let $\Pi^p(M)$ be the set of framed cobordism classes of codimension p in M. Define $\Pi^p(M) \times \Pi^q(M) \to \Pi^{p+q}(M)$ by transverse intersection, and for $p \geq \frac{1}{2}m + 1$ make $\Pi^p(M)$ an abelian group by disjoint union.

Product. Given framed (N, v) of codimension p and (N', w) of codimension q, deform N by Problem 16 to meet N' transversally. At $x \in N \cap N'$, transversality gives $T(N \cap N')_x = TN_x \cap TN'_x$ and $TM_x = TN_x + TN'_x$, so the normal space of $N \cap N'$, of dimension $p + q$, is spanned by the normal spaces of N and N' (the projection $TM_x \to TN_x^{\perp}$ restricted to TN'_x is onto, since $TN_x + TN'_x = TM_x$; project v and w onto the normal space of $N \cap N'$ and concatenate). This gives a framed submanifold of codimension $p + q$. Well defined on classes: a framed cobordism of N inside $M \times [0, 1]$, made transverse to $N' \times [0, 1]$ by the same argument, intersects it in a framed cobordism of the intersections; and two transverse deformations of N are joined by a framed cobordism (the isotopy $N_{(1-s)u+su'}$), which after a further transverse perturbation gives a cobordism between the two intersections.

Group structure for $p \geq \frac{1}{2}m + 1$. Then $n = m - p \leq \frac{1}{2}m - 1$, so $2n < m - 1$: two framed n-submanifolds can be made disjoint by a small isotopy (Problem 16 with \$N' = \$ the other one: transverse intersection has negative dimension, so is empty), and likewise two $(n + 1)$-dimensional cobordisms inside $M \times [0, 1]$, since $2(n+1) < m+1$. Define $[N] + [N'] := [N \sqcup N']$ after making them disjoint; well defined because disjoint cobordisms can be taken disjoint, and any two ways of making N, N' disjoint are connected through disjoint isotopies. Associativity and commutativity are clear (disjoint union is), the empty manifold is the identity, and the inverse of (N, v) is $(N, \bar{v})$ with the first framing vector negated: $N \sqcup N$ with framings v and $\bar{v}$ is the boundary of the framed cobordism obtained by pushing a copy of N into $M \times [0, 1]$ along a half-turn in the plane of v^1 and ∂_t, exactly as in Chapter 7, Claim 6.2(c). The result is the *cohomotopy group* $\pi^p(M)$; by Chapter 7 it is the set of homotopy classes of maps $M \to S^p$ with a group structure, and for $M = S^m$ it is the homotopy group $\pi_m(S^p)$. $\square$

Checked by machine. All seventeen problems have been verified in the Lean proof assistant. One value differs from the statement of Problem 15, as explained above: with the book's conventions the Hopf fibration has Hopf invariant -1. See the verified edition, §9, and *Milnor, proof by proof*, Parts VIII–XI.